

Backend Features – Presentation Bullet Points

1) User Management & Authentication

- Secure user registration and login
- Multiple roles: Patient, Doctor, Student Doctor, Receptionist, Admin
- Password hashing and JWT authentication
- Role-based authorization
- Account status control

2) Role-Based Access Control (RBAC)

- Role assignment at account creation
- Authorization policies per role
- Restricted access to sensitive endpoints
- Limited permissions for Student Doctors

3) Patient Management

- Patient profile and data management
- Secure medical records storage
- Medical and appointment history
- Controlled access to AI reports

4) Appointment Management

- Create, update, cancel appointments
- Reschedule and walk-in support
- Conflict prevention
- Appointment status tracking

5) Queue Management

- Patient check-in system
- Queue ordering and status updates
- Now-serving logic
- Concurrency handling

6) Doctor Dashboard Backend

- Daily and weekly schedules
- Patient medical records access
- Diagnosis and notes management
- AI report approval

7) Student Doctor Permissions

- View and edit AI reports
- No final diagnosis approval
- All actions logged

8) AI Diagnosis Integration

- X-ray image upload
- AI dental issue detection
- Confidence scores and overlays
- Doctor validation required

9) AI Report Management

- Version control
- Edit and approval tracking
- Secure report workflow

10) Payments & Financial Management

- Online and cash payments
- Payment status tracking
- Receipt generation
- Financial logs

11) Feedback & Reviews

- Patient feedback submission
- Doctor rating calculation
- Review validation

12) Notifications System

- Email, SMS, WhatsApp, In-app
- Appointment and payment alerts
- Delivery status logging

13) Administration

- User and role management
- Doctor verification
- System configuration
- Audit logs

14) Analytics & Reporting

- Appointments and revenue reports
- Doctor performance analytics
- AI accuracy metrics

15) Audit Logging & Compliance

- Authentication logs
- Medical record changes
- Financial transaction logs

16) Security & Data Protection

- HTTPS and JWT
- Input validation
- SQL injection prevention
- Secure file storage

17) Scalability

- Multi-clinic ready
- Modular architecture
- AI and system extensibility

18) Conclusion

- Secure and scalable backend
- Full clinic workflow management
- AI-assisted diagnosis with human control